

Neural Video Intelligence

How to present predicted brain-response proxies, behavioral outcomes, and real platform performance without turning the science into a fake "virality score".

TRIBE v2

Behavioral prediction

Cross-platform metrics

Audience

Frontend engineers, product designers, model/API engineers

Dataset context

~8k-10k unique short-form videos with Instagram, TikTok and YouTube metrics where available

Version

0.2 - 26 Sep 2026

[Changes from 0.1: see last page](#)

Goal

A compelling analysis UI that remains scientifically honest and useful for editors

What the interface is actually showing

The product should look like an advanced “brain + behavior” analysis system, but every visual must make clear which layer is measured, predicted, or merely a research proxy.

Observed

Real platform outcomes. Views, watch time, retention, shares, saves, comments, likes, completion, velocity and account context.

ground truth

Predicted behavior

Model outputs. Expected 3s/5s hold, completion, share rate, save rate, replay likelihood and normalized distribution potential.

trained prediction

Neural proxies

Research-derived latent signals. Attention, value, social relevance, cognitive friction, sensory drive and memorability proxies.

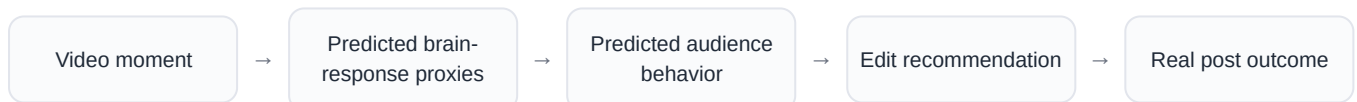
not literal intent

V0.2 DATA In the v0.2 data only the neural-proxy layer exists. `predictions.status` is `not_available` until a behavior model is trained on real outcomes; observed metrics arrive later from an exported outcome file.

Never write “brain firing” in the product. TRIBE v2 predicts cortical fMRI BOLD responses on an average-subject cortical surface. BOLD is a slow hemodynamic signal related to neural activity, not a direct recording of spikes. Use copy such as *predicted cortical response*, *neural proxy*, or *brain-response estimate*. [S1] When a proxy rises strongly, say “strong predicted response” (e.g. “Strong predicted attention capture”), not “firing”.

North-star interaction

The editor scrubs the video and sees what changes in the predicted audience response at that exact moment, then receives a concrete reason and an edit hypothesis. The timeline is the primary product; the 3D brain is the visual explainer.



In v0.2 the chain stops after the second box: neural proxies and rule-based moments are available; predicted audience behavior and real post outcomes are not yet connected.

Primary analysis view

The screen should answer three questions in under five seconds: **Where** does the video lose or gain attention? **Why?** **What** should I change?

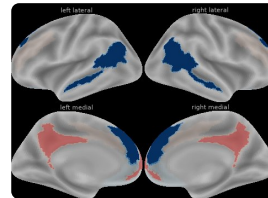
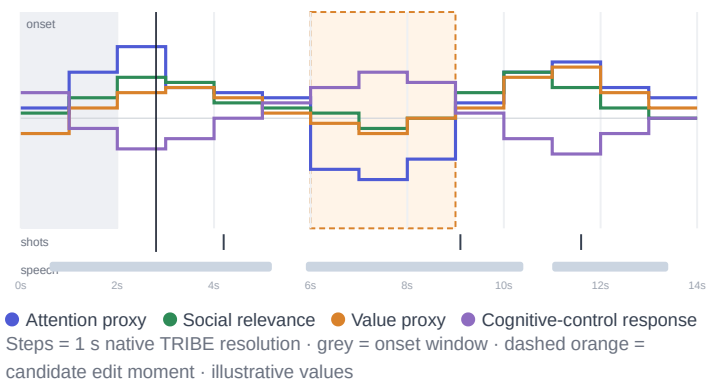
Neural Video Intelligence / Analysis a_dcd5bbec8ea7 · neural-tribe-roi-proxies_v0 · Research only · Confidence: uncalibrated

SYNTHETIC

VIDEO PLAYER - timeline synchronized

Predicted behavior: not available yet — behavior model not trained

predictions.status = not_available



t = 02.8s

● Attention proxy +1.4 z
● Social relevance +0.8 z
● Value proxy +0.5 z
● Cognitive-control response -0.6 z

TRIBE v2 prediction · average subject · research proxy, not measured brain activity

Moment insight

06.0 – 09.0 s · Attention proxy drop

Attention proxy falls 1.8 z below its level in the preceding 3 s while the shot stays unchanged, relative to this clip.

Untested edit hypothesis Test a visual state change here (cut, B-roll, zoom or caption change).

Compare edit

See evidence

Region	Purpose	Frontend rule	v0.2 binding
Top bar	Video/model identity, processing state, confidence	Always show model version and whether the result is research-only.	model_versions.neural, "Research only" (research_only is always true), confidence "uncalibrated"; red SYNTHETIC pill when synthetic: true.
Video player	Source of truth for time	Scrubbing must update every plot and brain map within one animation frame after data is loaded.	Time base is stimulus time; sample index = floor(t × timing.display_hz).
Outcome strip	Predicted behavioral metrics	Use probabilities/rates with intervals, not a universal score.	Must render the not_available state (empty card + predictions.reason) without placeholder numbers.
Timeline	Primary analysis tool	Allow toggling signals; default to 3-4 lines max to prevent chart noise.	Show channels[] with default_visible: true (4 lines); draw steps, keep gaps.
Brain map	Research explainer	Relative response only. Tooltip names ROI, proxy, z-score and evidence source.	Pre-rendered sprite tile for the current time (assets.brain_map); gap → dimmed + "no prediction".
Moment insight	Turns model output into action	Every recommendation must point to a timestamp and a measurable hypothesis.	moments[]: observations or untested edit hypotheses with a test_metric; no behavior deltas.

What should be displayed as a "good" neural response?

Do not encode one universal direction. Some useful content correlates involve increased activity; preferred short-video viewing has also been associated with **deactivation** of dACC/dlPFC relative to videos that viewers terminate early. The frontend therefore displays **pattern-specific proxies**, not “more = better”. [S3]

Frontend signal	Scientific basis	Display direction	Copy
Attention capture	Temporal reliability / synchrony; attention and sensory networks	Higher can be better, but compare against behavioral outcomes	“Strong predicted attention capture”
Sustained attention	Stable response through time; eventual product label learned from retention	Higher desired	“Attention remains stable through payoff”
Value / reward proxy	vmPFC/OFC cortical proxy; TRIBE checkpoint does not include NAcc	Higher hypothesized positive	“Higher cortical value proxy”
Social relevance	dmPFC / TPJ / precuneus networks associated with thinking about others and sharing	Higher hypothesized positive for transmission	“Higher social-relevance proxy”
Self relevance	mPFC / PCC-related self-processing	Context-dependent	“Self-relevance proxy rises here”
Cognitive friction	dACC / dlPFC control demand	Do not hard-code sign; learn from own dataset	“Control-network response changes”
Memorability	BOLD Moments labels + cortical correlates	Higher desired for recall, not necessarily watch time	“Predicted memorability”
Arousal / orienting	Future EEG/pupil student model from EMD	Context-dependent; peak timing matters	“Arousal proxy peak”

Important: neural synchrony is a better mental model for “the content is controlling attention” than raw activation. Prior naturalistic-media work linked temporal reliability across viewers to audience preferences and broad audience response. [S5]

Continued on the next page: the v0.2 channel keys that implement this taxonomy.

v0.2 channel keys

Generated from `tribe_research/brain/proxies_v0.yaml` (`proxies_v0`). Each channel is the vertex-weighted mean of its HCP-MMP1 ROI groups, z-scored within the clip. Labels and colours below are the exact strings the UI receives.

key	label	direction	default_visible	color	regions_text
attention	Attention proxy	hypothesis_positive	true	 #4F5BD5	Dorsal attention (FEF / IPS) and ventral attention / salience (insula / frontal operculum)
social	Social relevance	hypothesis_positive	true	 #2E8B57	TPJ / posterior STS and dmPFC
value	Value proxy	hypothesis_positive	true	 #D9822B	vmPFC / medial OFC (cortical proxy only; no NAcc in the TRIBE checkpoint)
control	Cognitive-control response	no_monotonic	true	 #8E6CC0	dACC and dlPFC
self	Self relevance	context_dependent	false	 #C0507A	mPFC, PCC and precuneus
language	Language / semantic load	context_dependent	false	 #3A8FB7	IFG (44/45), STG/STS language areas
sensory	Sensory drive	context_dependent	false	 #7A8A99	Early visual, visual motion (MT+) and auditory cortex

The direction enum

value	Frontend meaning
learned_positive	Higher is associated with better outcomes, validated on our held-out outcomes. No channel has this yet.
learned_negative	Lower is associated with better outcomes, validated on our held-out outcomes. No channel has this yet.
hypothesis_positive	Literature prior only: higher is hypothesized positive and must still be validated. Label it a hypothesis; no good/bad colouring.
context_dependent	Meaning depends on the content and the moment. Neutral colouring; describe, never grade.
no_monotonic	Neither direction is better; never colour one direction as good.

Mapping notes

- **Cognitive friction** ships as `control` ("Cognitive-control response") with `direction: no_monotonic`: the original "do not hard-code sign" rule, now machine-readable.
- **Sustained attention** is not its own channel. It is delivered as `channels[attention].summary.stability` ($1 / (1 + \text{sd of post-onset first differences})$).
- No channel is `learned_*` in v0.2, so no proxy may be shown as good or bad.

Timeline toggles

- ☒  Attention proxy
 ☒  Social relevance
☒  Value proxy
 ☒  Cognitive-control response
☐  Self relevance
 ☐  Language / semantic load
☐  Sensory drive

From `unavailable_channels`: disabled, with the reason text

- ☐ **Memorability** Needs a BOLD Moments student model (backend guide §06); not built.
☐ **Arousal** Needs an EEG/pupil student model (EEG Moments); not built.

The timeline is the core feature

Everything should resolve to timestamped curves and events. The editor needs to know **when** the video works, not merely whether the whole video scores well.

Required lanes → v0.2 data

Required lane (v0.1)	v0.2 source	Rendering
Observed retention / replay curve when available	<code>unavailable_channels</code> key <code>observed_retention</code> (kind <code>observed</code>)	Disabled lane with reason: "Outcome import not built yet."
Predicted hold / completion trajectory	<code>unavailable_channels</code> key <code>hold</code> (kind <code>behavior</code>)	Disabled lane with reason: "Behavior model not trained yet."
Attention, social relevance, value, cognitive-control response	<code>channels[]</code> with <code>default_visible: true</code>	Step lines in <code>color</code> ; <code>self</code> / <code>language</code> / <code>sensory</code> are off by default.
Shot boundaries	<code>events.shots_ms</code>	<code>[]</code> = computed, no cuts found; <code>null</code> = not computed (then <code>unavailable_lanes</code> contains <code>shots</code>).
Speech	<code>events.words</code> (<code>start_ms</code> , <code>end_ms</code> , <code>text</code>) and <code>events.speech_spans_ms</code>	Words closer than 0.6 s merge into one span. Empty in the sample (<code>quality.has_words: false</code>).
Text changes, speaker changes, audio onsets	<code>events.unavailable_lanes</code> (<code>text_changes</code> , <code>speaker_changes</code> , <code>audio_onsets</code>)	Disabled lanes with their reason; never guessed.
Model-generated candidate edit zones	<code>moments[]</code>	Observation = thin outline; <code>edit_hypothesis_untested</code> = dashed highlight.



Timing rules and moments

Timing rules

- **Grid.** Values sit on a 2 Hz display grid from $t = 0$ (`timing.display_hz`); `values[k]` covers $[k/2, (k+1)/2)$ s.
- **Steps.** The grid is filled by sample-and-hold from the native TRIBE step (`timing.native_tr_s`, 1.0 s in the sample). Draw steps; do not smooth or imply finer resolution.
- **Gaps.** `null` = no prediction (TRIBE drops segments with no events; spans in `timing.gaps_ms`). Leave a visible gap/hatch; never interpolate.
- **Onset.** `timing.onset_window_ms` [0, 2000] is the onset response. Shade it lightly and exclude it from “strongest moment” call-outs (`moment.in_onset_window`). The z scale is estimated outside it (`within_clip_z_excluding_onset`).
- **Time base.** `time_base`: “stimulus”: the 5 s hemodynamic lag is already removed, so t lines up with the video frame.

Moment rules (`moments.py`)

- `kind`: `attention_drop`, `broad_response` (≥ 3 channels rising together), `proxy_rise`, `proxy_fall`.
- Runs enter at $|z| \geq 1.0$, stay while $|z| \geq 0.5$, last ≥ 1.5 s; a gap ends a run. At most 12 moments.
- `attention_drop`: a fall run ≥ 1.5 z below the highest attention level in the preceding 3 s. Evidence adds `static_visual` (no shot boundary from 1 s before the run to its end), `speech_continues` (speech covers $\geq$ half the span), `language_load_high` (mean language $z > 0.5$).
- Only drops carry a `hypothesis` + `test_metric` (status `edit_hypothesis_untested`); everything else is an `observation`. `severity` ranks within this clip only.

Interaction

- Hover: crosshair on all lanes.
- Click: jump player and brain view to timestamp.
- Drag: select a segment and request “explain this span”.
- Shift-click: add segment to edit comparison.

MVP recommendation: ship a polished 2D cortical map or pre-rendered multi-view surface before investing in a heavy WebGL 3D brain. The timeline and edit explanation create the real product value.

v0.2 moment card

06.0 – 09.0 S

Untested edit hypothesis

Attention proxy drop

–1.8 z

vs the preceding 3 s (drop from its highest level there), relative to this clip

Evidence: `attention_drop` `static_visual`
`speech_continues`

Hypothesis (untested): Test a visual state change here (cut, B-roll, zoom or caption change).

Test metric: retention over this span (observed, or predicted once the behavior model exists)

`kind` `attention_drop` · `status` `edit_hypothesis_untested` · `confidence` `uncalibrated` · `in_onset_window` `false`. Illustrative values; no behavior delta is shown because v0.2 has none.

Target design

FUTURE · ONCE THE BEHAVIOR MODEL EXISTS ·
ILLUSTRATIVE V0.1 VALUE, NOT IN V0.2 DATA

02.4 – 03.8 S

–9 pts

predicted 5s hold impact vs local baseline

Pattern: attention falls while semantic load stays high and the visual scene is unchanged.

Hypothesis: payoff arrives too late.

Action: bring the reveal forward by ~3.5s or introduce a visual state change.

The language should remain an edit hypothesis until it is validated by real posting outcomes.

Behavioral predictions must be separate from neural proxies

v0.2: `predictions = {status: "not_available", reason, metrics: {}}`. Render an empty state with `reason`; never show placeholder numbers. The schema forbids metrics while the status is `not_available`.

PREDICTED BEHAVIOR

Behavior model not trained yet.

Hold, completion and share predictions appear here once a behavior model has been trained and calibrated on real outcomes.

Target design

TARGET DESIGN · ILLUSTRATIVE V0.1 VALUES · NOT IN V0.2 DATA

3S HOLD

81%

76-85% interval

5S HOLD

74%

69-79% interval

COMPLETION

61%

duration-normalized

SHARE RATE

P74

creator-normalized percentile

Recommended grouping

Retention

1s/3s/5s hold, average watch %, completion, predicted drop zones.

Transmission

Shares/sends, saves, replay likelihood, social-relevance contribution.

Distribution

Normalized expected views, velocity and platform-specific expansion potential.

Avoid "viral probability" in the first version. Platform distribution is a feedback process influenced by account, exposure, timing and early feedback. Show normalized predictions per metric, plus uncertainty. Add a single headline only after the model has been calibrated on real held-out posts.

Cross-platform view

TARGET DESIGN · ILLUSTRATIVE V0.1 VALUES · NOT IN V0.2 DATA

Metric	TikTok	Instagram	YouTube Shorts	Intrinsic content estimate
5s hold	76%	71%	79%	75%
Completion	58%	62%	65%	61%
Share percentile	P82	P88	P66	P79
Distribution residual	+0.9	+0.5	+0.3	+0.6

Values above are illustrative. "Intrinsic" should come from a hierarchical content-quality model, not a simple average.

Variant comparison should be a first-class workflow

The product becomes much more useful when an editor can compare the original cut against proposed revisions before publishing.

TARGET DESIGN · ILLUSTRATIVE V0.1 VALUES · NOT IN V0.2 DATA

Original	Edit 1	Edit 2
A	B	C
5s hold: 68%	5s hold: 76% +8	5s hold: 72% +4
Completion: 54%	Completion: 58% +4	Completion: 63% +9
Share percentile: P61	Share percentile: P64	Share percentile: P75 +14
Largest risk: 2.1-4.4s payoff delay.	Reveal moved to 2.7s.	Shorter intro + social framing.

V0.2 DATA Behavior deltas require the behavior model (P1). Until then a comparison can only place the neural lanes of two analyses side by side, and each stays relative to its own clip: within-clip z-values are not comparable across clips, so no neural delta should be computed either.

Diff UX

- Align variants by semantic moment as well as timestamp when cuts change duration.
- Show deltas in predicted behavior first; neural differences second, as which moments and channels differ between cuts, never as z-value deltas (each clip is normalised on its own).
- Each delta must show confidence and model version.
- Allow “publish as experiment” to create a backend experiment object and later attach real outcomes.

Long-term loop: recommendation → edited variant → real post → measured uplift → training example. That feedback loop is the defensible asset; the brain visualization is a feature layer, not the moat.

Frontend data contract

The UI should not know how TRIBE, EEG proxies or behavior models are implemented. It consumes a versioned analysis object with aligned time-series channels. v0.2 freezes that object as `nvi.analysis.v0.2:docs/analysis.schema.json` (JSON Schema 2020-12), with a synthetic example bundle in `docs/sample_analysis/`.

GET /v1/analyses/{analysis_id} excerpt of docs/sample_analysis/cd20b16879d630c4/analysis.json (synthetic)

```
{
  "schema_version": "nvi.analysis.v0.2",
  "analysis_id": "a_dcd5bbec8ea7",
  "video_id": "cd20b16879d630c4",
  "source_name": "clip",
  "duration_ms": 26100,
  "status": "complete",
  "research_only": true,
  "synthetic": true,
  "model_versions": {"neural": "tribe-roi-proxies_v0", "behavior": null, "explain": "neural-moments-0.1.0"},
  "provenance": {"tribe_commit": null, "roi_map": {...}, "proxies_version": "proxies_v0",
    "feature_version": "summary_v0",
    "normalization": "within_clip_z_excluding_onset"},
  "timing": {
    "time_base": "stimulus",
    "note": "TRIBE predictions are pre-shifted 5 s for hemodynamic lag; t=0 is the first video frame.",
    "native_tr_s": 1.0,
    "display_hz": 2,
    "n_display_samples": 53,
    "onset_window_ms": [0, 2000],
    "gaps_ms": []
  },
  "predictions": {"status": "not_available", "reason": "Behavior model not trained yet.", "metrics": {}},
  "channels": [
    {
      "key": "attention",
      "label": "Attention proxy",
      "kind": "neural_proxy",
      "unit": "z_within_clip",
      "direction": "hypothesis_positive",
      "direction_note": "Higher can be better, but compare against behavioral outcomes before relying on it.",
      "confidence": "uncalibrated",
      "source_model": "tribev2",
      "research_status": "research_proxy",
      "hz": 2,
      "native_tr_s": 1.0,
      "resampling": "sample_and_hold",
      "default_visible": true,
      "color": "#4F5BD5",
      "basis": {
        "roi_groups": ["attention_dorsal", "attention_ventral"],
        "regions_text": "Dorsal attention (FEF / IPS) and ventral attention / salience (insula / frontal operculum)",
        "atlas": "HCP-MMP1"
      },
      "copy": {
        "tooltip": "Predicted response in attention and salience networks, relative to the rest of this clip.",
        "rise": "Strong predicted attention capture",
        "fall": "Predicted attention response falls"
      },
      "values": [-1.258, -1.258, -2.242, -2.242, 0.453, ... (53 total)],
      "summary": {
        "mean_0_3s": -1.016,
        "mean_0_5s": -0.677,
        "peak_z": 2.06,
        "peak_time_ms": 23000,
        "trough_z": -2.242,
        "trough_time_ms": 1000,
        "stability": 0.396
      }
    },
    ... 6 more: social, value, control, self, language, sensory
  ],
  ... continued
}
```

```

...
"unavailable_channels": [
  {
    "key": "memorability",
    "kind": "neural_proxy",
    "reason": "Needs a BOLD Moments student model (backend guide §06); not built."
  },
  {
    "key": "arousal",
    "kind": "neural_proxy",
    "reason": "Needs an EEG/pupil student model (EEG Moments); not built."
  },
  {"key": "hold", "kind": "behavior", "reason": "Behavior model not trained yet."},
  {"key": "observed_retention", "kind": "observed", "reason": "Outcome import not built yet."}
],
"events": {
  "shots_ms": [],
  "words": [],
  "speech_spans_ms": [],
  "unavailable_lanes": [
    {"key": "text_changes", "reason": "No OCR pass yet."},
    {"key": "speaker_changes", "reason": "No diarization pass yet."},
    {"key": "audio_onsets", "reason": "No audio-onset detector yet."}
  ]
},
"moments": [
  ... m1-m3 (observations),
  {
    "kind": "attention_drop",
    "start_ms": 6000,
    "end_ms": 9000,
    "channels": ["attention"],
    "peak_z": -1.45,
    "severity": 0.97,
    "title": "Attention proxy drop",
    "description": "Attention proxy falls while the shot stays unchanged. Drop of 2.9 z, relative to this clip.",
    "evidence": ["attention_drop", "static_visual"],
    "hypothesis": "Test a visual state change here (cut, B-roll, zoom or caption change).",
    "test_metric": "retention over this span (observed, or predicted once the behavior model exists)",
    "status": "edit_hypothesis_untested",
    "relative_to": "this_clip",
    "confidence": "uncalibrated",
    "in_onset_window": false,
    "id": "m4"
  },
  ... m5-m12 (12 total)
],
"assets": {
  "region_map": {
    "idmap_src": "../_static/region_idmap.png",
    "legend_src": "../_static/region_legend.png",
    "ids": {"#4f5bd5": "attention", "#2e8b57": "social", ... 7 total}
  },
  "brain_map": {
    "mode": "proxy",
    "src": "brain_proxy.jpg",
    "tile_w": 480,
    "tile_h": 360,
    "cols": 10,
    "rows": 3,
    "frame_start_ms": [0, 1000, 2000, 3000, 4000, ... (27 total)],
    "frame_end_ms": [1000, 2000, 3000, 4000, 5000, ... (27 total)],
    "views": ["left lateral", "right lateral", "left medial", "right medial"],
    "colormap": {"name": "RdBu_r", "vmin": -2.5, "vmax": 2.5, "threshold": 0.5, "unit": "z_within_clip"}
  }
},
"quality": {
  "n_segments": 27,
  "has_words": false,
  "warnings": [
    "SYNTHETIC: dry-run predictions, not TRIBE output.",
    "No transcribed words: language-driven signals rely on audio/video only."
  ]
}
}

```

Signal metadata and contract rules

Top-level keys

Key	Frontend use
schema_version	"nvi.analysis.v0.2"; refuse unknown versions.
analysis_id, video_id, source_name	Identity; source_name is the upstream id for the later outcome join.
duration_ms, status	Timeline length; status is queued, processing, complete or failed.
research_only, synthetic	Always true / SYNTHETIC banner when true.
model_versions, provenance	Top bar and "See evidence"; behavior is null in v0.2.
timing	Grid, native step, onset window and gaps (§04 cont.).
predictions	not_available in v0.2: empty state (§05).
channels, unavailable_channels	Timeline lanes and toggles (§03 cont., §04).
events, moments	Event lanes and candidate zones (§04).
assets	Brain map sprite, region map, legend (§08).
quality	warnings render as notices next to the top bar.

Required metadata on every signal

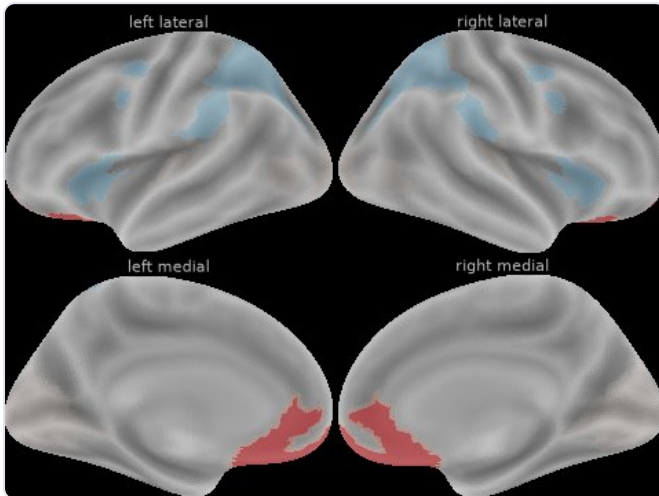
v0.1 requirement	v0.2 field(s)	Value in v0.2 data
kind	kind	"neural_proxy" (enum: neural_proxy, behavior, observed)
unit	unit	"z_within_clip" (constant)
direction	direction + direction_note	enum in §03 (cont.); note is display copy
confidence	confidence	"uncalibrated" (enum: uncalibrated, low, medium, high)
source model	source_model (+ model_versions.neural)	"tribev2" / "tribe-roi-proxies_v0"
time resolution	hz, native_tr_s, resampling	2, 1.0, "sample_and_hold"
research status	research_status	"research_proxy" (enum: research_proxy, validated)

Never infer semantics from a channel name on the frontend. The backend must explicitly state whether higher/lower has a learned positive association, a hypothesis-only direction, or no monotonic interpretation.

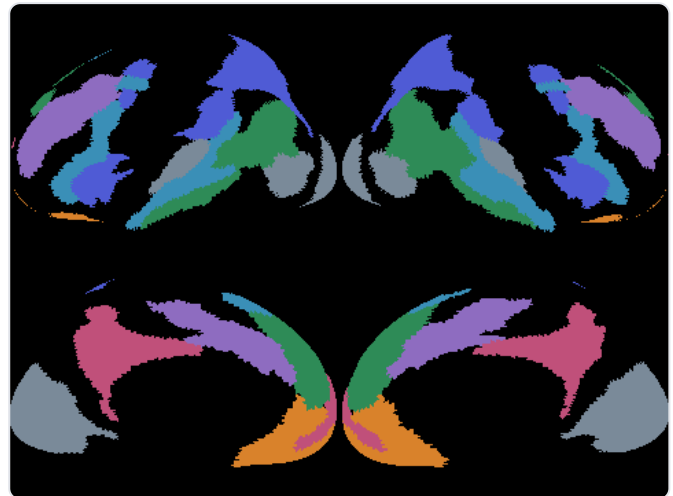
Contract rules. All values are within-clip z-scores: never compare them across clips. severity ranks moments within this clip only. synthetic: true must show a SYNTHETIC banner. research_only is always true. Asset paths are relative to the analysis.json folder.

Pre-rendered cortical views, synchronized to time

The brain is rendered offline, so the frontend only swaps sprite tiles and reads pixels. It needs no WebGL.



`assets.brain_map`, tile 0 (0–1000 ms, inside the onset window). Synthetic sample — dry-run predictions, real HCP-MMP1 regions.



`assets.region_map.idmap_src`: flat colour per channel, same tile geometry. Synthetic sample — dry-run predictions, real HCP-MMP1 regions.

Sprite field	Meaning
<code>mode</code>	"proxy": each channel's region painted with its z. "research_vertex": per-vertex map.
<code>src</code>	Sprite image, relative to the analysis.json folder (<code>brain_proxy.jpg</code>).
<code>tile_w, tile_h</code>	Tile size in px (480 × 360). Each tile holds 4 views in a 2×2 layout.
<code>cols, rows</code>	Sprite grid, row-major (10 × 3 in the sample; 27 tiles used).
<code>frame_start_ms, frame_end_ms</code>	One entry per tile = one native TRIBE step. Show tile <i>i</i> while <code>frame_start_ms[i] ≤ t < frame_end_ms[i]</code> .
<code>views</code>	left lateral, right lateral, left medial, right medial.
<code>colormap</code>	RdBu_r diverging, vmin -2.5 / vmax 2.5, threshold 0.5 (faded below), unit <code>z_within_clip</code> .

Region hover (`assets.region_map`)

1. Map the cursor position inside the current brain tile to `idmap` pixel coordinates (scale by `tile_w/tile_h`).
2. Sample the lossless `idmap` PNG at native size with smoothing off (never the JPEG sprite); format as lowercase `#rrggb`.
3. Look it up in `ids` → channel key. `#000000` = no channel: no tooltip.
4. Tooltip: `label`, `basis.regions_text`, current `z` (`values[floor(t × display_hz)]`) and `copy.tooltip`.

The `idmap` is static per ROI-map version. `legend_src` shows channel colours, directions and the colormap.

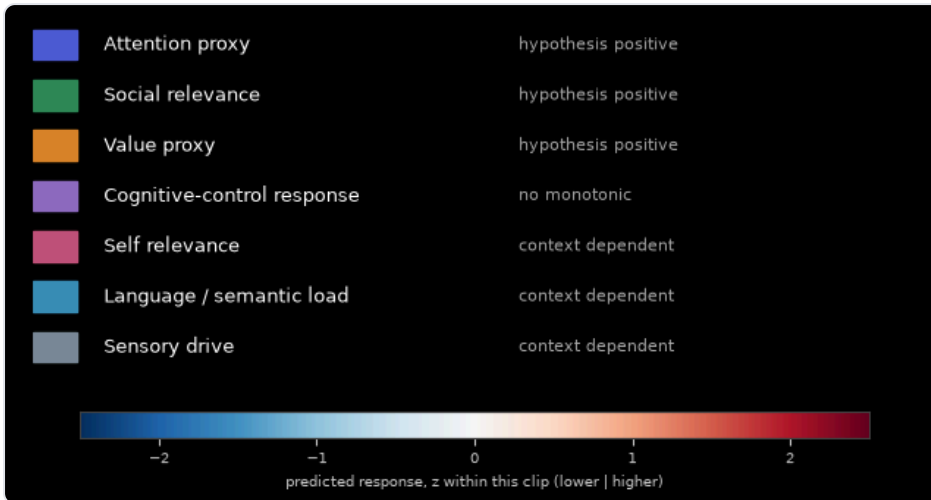
● Attention proxy +1.4 z relative to this clip

Dorsal attention (FEF / IPS) and ventral attention / salience (insula / frontal operculum)

Predicted response in attention and salience networks, relative to the rest of this clip.

Research proxy · uncalibrated

Always caption the brain: **TRIBE v2 prediction · average subject · research proxy, not measured brain activity**



`assets.region_map.legend_src`: channel colours, `direction` and the `RdBu_r` scale. Synthetic sample — dry-run predictions, real HCP-MMP1 regions.

Gaps and onset

If `t` falls in `timing.gaps_ms` (no tile covers it), dim the brain panel and label it “no prediction”. Never hold the previous tile silently. Tiles inside the onset window get the same light “onset response” note as the timeline.

`assets.research_vertex_map`

Same sprite shape with `mode: "research_vertex"` (`brain_vertex.jpg` in real bundles): the raw per-vertex prediction. Show it only when Research mode is enabled. Absent from the synthetic sample. Optional `summary_png` and `demo_mp4` are also absent there.

Colour is relative. Red/blue shows a region above/below its own average in this clip. It does not mean good/bad and is not comparable across clips; regions below the 0.5 z threshold are faded.

States, confidence and wording

State	UI treatment	Recommended wording
Research-only neural signal	Orange “Research proxy” badge	“Predicted social-relevance response”
Behavior model calibrated	Blue metric + interval	“Predicted 5-second hold: 74% (69-79%)” illustrative, not v0.2
Low confidence / OOD	Muted chart, warning icon	“Low confidence: this format is underrepresented in training data.”
Observed post data	Green “Observed” badge	“Actual 5-second hold: 78%” illustrative, not v0.2
Edit hypothesis	Dashed highlight	“Test moving the reveal earlier.”
Validated edit uplift	Solid highlight + experiment ID	“Variant B improved normalized 5s hold by +6.1 pts.” illustrative, not v0.2
v0.2 SYNTHETIC data	SYNTHETIC red banner when <code>synthetic: true</code>	“SYNTHETIC — dry-run data, not model output”
v0.2 Behavior not available	Empty card, no numbers	“Behavior model not trained yet”
v0.2 No prediction gap	Hatched/dimmed lane; brain dimmed	“no prediction”
v0.2 Onset window	Muted shading over 0–2 s	“first 2 s: onset response”
v0.2 Candidate moment (observation)	Thin outline	“Observation · relative to this clip”
v0.2 Edit hypothesis, untested	Dashed highlight	“Untested edit hypothesis”
v0.2 Uncalibrated confidence	Muted confidence label	“Uncalibrated research proxy”

Copy rules

- Prefer “associated with” and “proxy for” over “means”.
- Never say a specific cortical region “caused” a viewer to share.
- Never claim the model reads a specific viewer’s mind.
- Do not expose raw 20,484-vertex values unless “Research mode” is enabled.
- Keep editor-facing language behavioral: hook, payoff, pacing, novelty, social value, friction.
- v0.2 Say “strong predicted response”, never “brain firing”.
- v0.2 Every z-value is relative to this clip.
- v0.2 Never show a proxy as good/bad unless its direction is `learned_positive` / `learned_negative`.

Frontend MVP build order

Priority	Component	Why	Data available in v0.2
P0	Video player + synchronized multi-signal timeline	Core analysis interaction.	Yes: neural lanes, shots, speech
P0	Behavior prediction cards + confidence	Connects analysis to measurable outcomes.	No: render the not_available empty state
P0	Moment inspector + edit hypothesis	Turns model signal into action.	Yes: moments[]
P0	2D cortical map synchronized to timeline	Strong demo value with manageable complexity.	Yes: brain map sprite + region map
P1	Original vs edit comparison	Creates iterative workflow.	No: deltas need the behavior model
P1	Observed outcome overlay	Closes model-to-reality loop.	No: outcome import not built
P2	3D WebGL brain	Great showcase; limited incremental utility.	No: 2D pre-rendered views only
P2	Cross-platform intrinsic-quality panel	Requires backend hierarchical model to be trustworthy.	No
Avoid	Universal 0-100 "viral brain score"	Invites overclaiming and hides metric-specific signal.	—

Definition of a good demo: a creator can play a clip, scrub to a weak moment, see exactly which predicted signals changed, receive a concrete edit hypothesis, compare a revised cut, and later overlay the real platform outcome.

V0.2 DATA A v0.2 demo can cover the first four steps (play, scrub, see which proxies changed, read an untested edit hypothesis). Comparing a revised cut on behavior and overlaying real outcomes wait for the behavior model and the outcome import.

Research references used for the UI taxonomy

1. **[S1] TRIBE v2.** d'Ascoli et al. (2026), *A foundation model of vision, audition, and language for in-silico neuroscience*. Official repository describes multimodal prediction onto the fsaverage5 cortical surface (~20k vertices). github.com/facebookresearch/tribev2
2. **[S2] BOLD Moments Dataset.** 1,102 three-second videos, fMRI responses from 10 subjects, rich labels including memorability. [Nature Communications \(2024\)](#); [dataset repo](#).
3. **[S3] Preferred short-video viewing.** Hong et al. (2026), *Brain activity inhibition during Short Video Viewing*. Preferred videos watched to completion showed greater dACC/dlPFC deactivation than less-preferred videos terminated early. [doi:10.1016/j.neuroimage.2026.121722](https://doi.org/10.1016/j.neuroimage.2026.121722)
4. **[S4] Sharing and social relevance.** Scholz et al., *The Value of Sharing Information*. Value, self-related and social-cognition systems were associated with selection/sharing; dmPFC/TPJ showed sharing-specific associations in whole-brain analyses. [PMC5507728](#)
5. **[S5] Neural synchrony and audience preference.** Dmochowski et al. (2014), *Audience preferences are predicted by temporal reliability of neural processing*. [Nature Communications 5:4567](#)
6. **[S6] EEG Moments Dataset.** 1,102 three-second videos, 128-channel EEG and eye tracking from 6 subjects; same stimuli as BOLD Moments. [arXiv:2608.28768](#); [OpenNeuro mirror](#).
7. **[S7] SmartEdit.** Gupta et al. (ICME 2025), edit-signal-based short-video engagement prediction and improvement suggestions. [Adobe Research](#).
8. **[S8] SnapUGC / VQualIA.** Large real-world short-video engagement dataset and ICCV 2025 challenge; engagement models benefit from multimodal video/audio/text features. [ICCV Open Access](#).
9. **[S9] HCP-MMP1 atlas.** Glasser et al. (2016), *A multi-modal parcellation of human cerebral cortex*. *Nature* 536:171–178. fsaverage projection from figshare 3498446, mirrored at github.com/tannerjared/HCP-MMP1.

Scientific interpretations here are design priors, not validated biomarkers of virality. Product claims should be calibrated only after held-out evaluation on the company's own platform outcomes.

Changelog 0.1 → 0.2

- **Cover** Version 0.2 - 26 Sep 2026, with a pointer to this page.
- **§01** Blue v0.2 note: only the neural-proxy layer exists and `predictions.status` is `not_available`. Terminology callout adds “strong predicted response” wording.
- **§02** Mock rebuilt for v0.2 data: honest top bar with SYNTHETIC pill, exact proxy labels, sprite brain tile, behavior empty state, v0.2 moment. Region table adds a “v0.2 binding” column.
- **§03** New continuation page: v0.2 channel keys from `proxies_v0.yaml`, the `direction` enum, sustained attention as `summary.stability`, memorability/arousal as disabled toggles.
- **§04** Required lanes mapped to fields, with a new lane figure. Timing rules: 2 Hz grid, sample-and-hold steps, null gaps, onset window, stimulus time. New moment-rules summary. v0.2 moment card with no behavior delta; the -9 pts card is kept only as a labelled future example.
- **§05** Orange callout and empty state for `predictions: not_available`. The v0.1 numbers are labelled illustrative target design.
- **§06** Variant numbers labelled illustrative. Note: deltas require the behavior model (P1); within-clip z-values are not comparable across clips, so neural differences are shown as differing moments/channels, never z deltas.
- **§07** Now three pages. Example replaced by an excerpt of the real `nvi.analysis.v0.2` sample. New top-level keys table. Signal metadata mapped to field names, plus contract rules.
- **§08 (new)** Brain map assets: `brain_map` sprite, `research_vertex_map`, `region_map` hover lookup, legend, required brain caption.
- **§09** Renumbered from 08. Seven new v0.2 UI states and three new copy rules. v0.1 example numbers labelled illustrative.
- **§10** Renumbered from 09. Added a “Data available in v0.2” column and a v0.2 demo note.
- **§11** Renumbered from 10. Added [S9] Glasser et al. (2016), HCP-MMP1.

Data contract: `docs/analysis.schema.json` (`nvi.analysis.v0.2`, frozen). Example: `docs/sample_analysis/` (synthetic).